

Problem Statement

Evaluate the limit from the image "dailyintegral-limit-limit-1x1-2026-07-29.jpg":

$$L = \lim_{x \rightarrow \infty} \left(\frac{1}{\sqrt{x^2 + 1}} + \frac{1}{\sqrt{x^2 + 2}} + \cdots + \frac{1}{\sqrt{x^2 + x}} \right)$$

Detailed Solution

To evaluate the limit L as $x \rightarrow \infty$, we can express the expression inside the limit as a finite sum of x terms and apply the **Squeeze Theorem** (also known as the Sandwich Theorem).

Step 1: Expressing the Sum and Setting Up Inequalities

Let $S(x)$ denote the sum of the x terms for a given positive integer x :

$$S(x) = \sum_{k=1}^x \frac{1}{\sqrt{x^2 + k}}$$

For every index $k \in \{1, 2, \dots, x\}$, the denominator $\sqrt{x^2 + k}$ satisfies the inequality:

$$\sqrt{x^2 + 1} \leq \sqrt{x^2 + k} \leq \sqrt{x^2 + x}$$

Taking the reciprocal reverses the inequalities:

$$\frac{1}{\sqrt{x^2 + x}} \leq \frac{1}{\sqrt{x^2 + k}} \leq \frac{1}{\sqrt{x^2 + 1}}$$

Step 2: Bounding the Sum

Summing these inequalities over all x terms from $k = 1$ to $k = x$, we obtain lower and upper bounds for $S(x)$:

$$\sum_{k=1}^x \frac{1}{\sqrt{x^2 + x}} \leq \sum_{k=1}^x \frac{1}{\sqrt{x^2 + k}} \leq \sum_{k=1}^x \frac{1}{\sqrt{x^2 + 1}}$$

Since the bounding terms do not depend on the summation index k , adding x identical terms simplifies to multiplying by x :

$$\frac{x}{\sqrt{x^2 + x}} \leq S(x) \leq \frac{x}{\sqrt{x^2 + 1}}$$

Step 3: Evaluating the Limits of the Bounds

We now compute the limit as $x \rightarrow \infty$ for both the lower and upper bounds by dividing the numerator and denominator by $x = \sqrt{x^2}$ (for $x > 0$):

Lower Bound Limit:

$$\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + x}} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{x}}} = \frac{1}{\sqrt{1 + 0}} = 1$$

Upper Bound Limit:

$$\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + 1}} = \lim_{x \rightarrow \infty} \frac{1}{\sqrt{1 + \frac{1}{x^2}}} = \frac{1}{\sqrt{1 + 0}} = 1$$

Step 4: Applying the Squeeze Theorem

Since both the lower and upper bounds converge to the same value:

$$\lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + x}} = 1 \quad \text{and} \quad \lim_{x \rightarrow \infty} \frac{x}{\sqrt{x^2 + 1}} = 1$$

by the Squeeze Theorem, the limit of $S(x)$ as $x \rightarrow \infty$ must also be 1:

$$L = \lim_{x \rightarrow \infty} S(x) = 1$$

Final Answer:

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